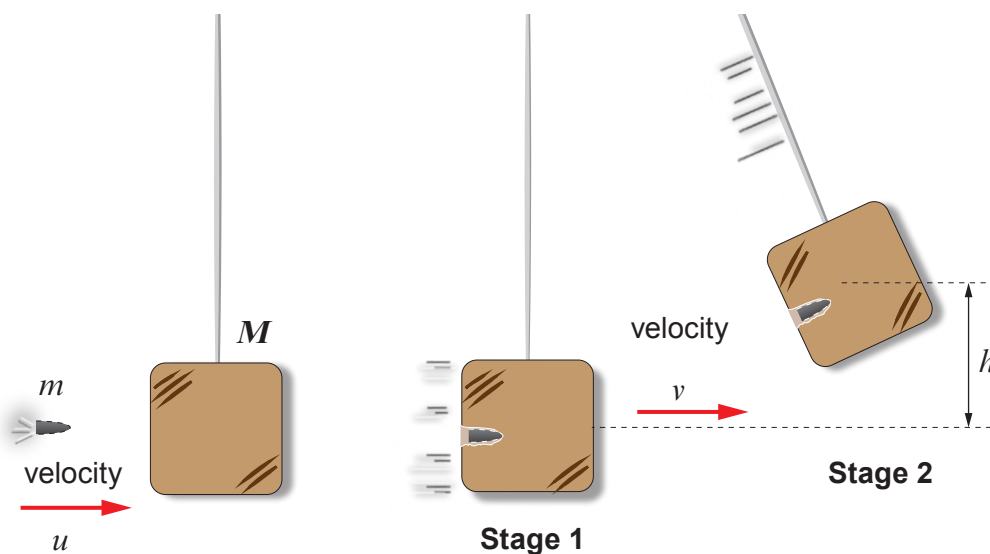


7. The velocity of a bullet can be determined by firing it into a suspended wooden block. As the bullet enters the block it will cause it to swing through a height, h . This is shown in the diagram below.



- (a) (i) State the law of conservation of momentum. [2]

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- (ii) Explain whether this law applies to the block as it swings up. [1]

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- (b) Discuss whether kinetic energy is conserved in **both** stages of the process. [2]

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- (c) The mass of a bullet is 16 g. It is fired at a velocity, u into a stationary wooden block of mass 2.5 kg. The combined bullet and block start to move with a velocity, v . The block swings through a height, h , equal to 11.9 cm before it comes to rest again.

Use the information above and equations from page 2 to answer the following questions.
($g = 10 \text{ m/s}^2$)

- (i) Calculate the velocity, v . [4]

Velocity, $v = \dots\dots\dots \text{ m/s}$

- (ii) Use your answer in part (i) to calculate the velocity, u , of the bullet. [4]

Velocity, $u = \dots\dots\dots \text{ m/s}$

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END OF PAPER

